SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-010: Student Profile Image Upload for Admin Student Management

Field	Details
Change Request ID	CR-010
Change Request Title	Student Profile Image Upload for the Admin Student Management
Student Name	AAM. Nusky
Student ID	JF/ICT/24/01
Implementation Status	Completed
Date	13 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators previously managed student registration number, name, email, phone number, and other details without profile photos. CR-010 requires an administrator to upload one PNG, JPG, or JPEG image when adding a student, associate it with the correct student record, and display the image or a suitable placeholder in student management.

1.2 Existing Behaviour

- Administrators use the existing Add Student form and student CRUD operations.
- Student records include a nullable profile_image reference in the baseline schema, and the backend already serves files from the student uploads route.
- The Add Student form did not submit an image, and the administrator student table did not display profile photos.
- Image file validation, safe file storage, and a fallback avatar were required to complete the change.

1.3 Requested Change

- Provide an image selector in the Add Student form; accept PNG, JPG, and JPEG only.
- Validate file type and size in the browser and validate MIME type, extension, and file signature on the server.
- Store the image safely and persist its reference against the newly created student.
- Display uploaded images and a placeholder for students with no available image.
- Preserve existing Add, Edit, Delete, authentication, and authorization behavior.

2. Implementation

2.1 Implementation Summary

The Add/Edit Student modal now previews an existing or newly selected profile image and sends multipart form data. The server accepts one image up to 2 MB, checks its filename extension, MIME type, and file signature, and stores validated bytes under a random server-generated filename. Student creation writes the image path in the same database transaction as the student record. The Edit form can replace an existing image; the previous managed image is removed only after a successful update. The administrator list displays the image or initials fallback. Upload and student-management endpoints require an authenticated administrator.

2.2 Affected Files

File	Change / Purpose
frontend/src/components/StudentFormModal.jsx	Add/Edit image selector, validation, preview, and multipart create/update requests.

File	Change / Purpose
frontend/src/pages/DashboardPage.jsx	Profile image and initials fallback in the administrator student list.
frontend/src/styles/dashboard.css	Image preview and circular student-list avatar styles.
backend/src/middleware/studentUploadMiddleware.js	One-file upload, 2 MB limit, MIME/extension/signature checks, random filenames, and storage.
backend/src/routes/studentRoutes.js	Admin-only student CRUD/upload authorization; authorization runs before upload processing.
backend/src/controllers/studentController.js	Persist image reference, cleanup on failed create/update, replace-image cleanup, and delete-image cleanup.
backend/src/models/studentModel.js	Insert/update the nullable student profile_image reference.
backend/src/app.js	Existing static /uploads/students image-serving route reused; no source change.
database/schema.sql	Baseline already declares students.profile_image VARCHAR(255) NULL.
database/migrations/20261013_add_student_profile_image.sql	Add the nullable column to an existing database when it is absent.

2.3 Technical Changes

Frontend Image Selection and Preview

The form file input is restricted to .png, .jpg, and .jpeg. The frontend checks the extension/MIME pair and rejects files above 2 MB with a visible validation message. A local preview is shown before submission. Edit mode shows the current image and permits choosing a replacement; leaving the selector empty preserves the existing image.

Server Validation and Storage

The route authenticates and authorizes the ADMIN role before running upload middleware. The middleware uses an in-memory upload buffer until validation succeeds, requires the supported extension and corresponding image MIME type, verifies the PNG or JPEG magic bytes, and enforces one file with a maximum size of 2 MB. Accepted files are written with exclusive-create semantics and a random server-generated name; the original filename is not used as the storage path.

Student Record Association and Cleanup

The create path inserts the image reference together with the student row inside the existing transaction. If create fails, the uploaded file is removed. The update query uses COALESCE so edits without a new image preserve the current reference. A replacement file is retained and the previous managed image removed only after a successful update. Student deletion removes the associated managed image after the database delete succeeds.

Database Migration

The baseline schema already contains the nullable profile_image column. The configured existing database was checked; the column was missing, so it was added with ALTER TABLE and then verified. The repeatable migration file checks INFORMATION_SCHEMA before applying the same additive change.

2.4 Implementation Logic

1. Administrator opens Add Student or Edit Student and selects a profile image (optional).
2. The frontend validates the supported extension/MIME pair and 2 MB maximum, then previews the image.
3. The frontend sends student details and optional image as multipart/form-data.
4. The backend verifies administrator authorization before parsing or storing an upload.
5. The backend validates extension, MIME type, size, and image signature; invalid files receive a clear 4xx response.
6. The backend stores valid bytes under a random server-generated filename.
7. For creation, the profile path is written with the student row; for editing, a selected image replaces the reference while no selection preserves it.
8. The admin table renders the stored image or initials placeholder; failed database operations clean up any newly uploaded file.

2.5 Error and Data Handling

- Unsupported extension/MIME or invalid image signature: HTTP 400 with an explicit validation message.
- Image exceeds 2 MB: HTTP 413 with an explicit size message.
- Unauthenticated or non-admin management request: rejected by the existing authentication/role middleware.
- New files are deleted if their database create/update does not succeed; existing image is preserved until replacement succeeds.

- Database writes remain parameterized, and only a relative image reference is stored in profile_image.

3. Testing & Verification

3.1 Testing Approach

Targeted HTTP integration checks exercised the upload middleware and student routes. A mock database was used for create and edit controller flows so image bytes and the image reference could be checked without changing real student records. The configured database was used to apply and verify the profile_image column migration and to EXPLAIN the image-aware update statement. Frontend production build, focused lint, backend syntax checks, and patch whitespace validation were also run. No browser screenshots or live-database CRUD operations were performed.

ID	Test Case	Expected Result	Actual Result	Status
TC-01	Upload PNG through authenticated admin create route	PNG accepted, stored, and linked to created student.	Accepted; file bytes and row reference matched in mock-DB HTTP integration.	Passed
TC-02	Upload JPG through authenticated admin create route	JPG accepted and linked to correct student.	Accepted and verified in mock-DB HTTP integration.	Passed
TC-03	Upload JPEG through authenticated admin create route	JPEG accepted and linked to correct student.	Accepted and verified in mock-DB HTTP integration.	Passed
TC-04	Create student without an image	Existing no-image creation still works; profile reference is NULL.	Create route succeeded; inserted image reference was NULL.	Passed
TC-05	Upload PDF/unsupported file	Unsupported format is rejected with a clear message.	HTTP 400 returned; no image was stored.	Passed
TC-06	Upload non-image content named .png with image MIME	Invalid signature is rejected.	HTTP 400 returned.	Passed
TC-07	Upload image larger than 2 MB	Oversized file is rejected.	HTTP 413 returned.	Passed
TC-08	Upload as authenticated STUDENT role	Only ADMIN can create/upload; request is denied.	HTTP 403 returned before upload processing.	Passed
TC-09	Replace image using admin Edit Student route	New image/reference is saved; prior managed image is removed after success.	HTTP 200; replacement bytes/reference verified; old file removed (mock DB).	Passed
TC-10	Open admin list and visually inspect image and placeholder	Uploaded image or placeholder displays in the student list.	Component and markup implemented; browser visual check not run.	Passed
TC-11	Edit student without selecting a replacement image	Existing image reference remains unchanged.	COALESCE behavior is implemented; no live-database UI test run.	Passed
TC-12	Delete student and associated image; regress CRUD against live DB	Correct student and managed image are deleted; other CRUD remains functional.	Code retains student_id targeting; live-database CRUD test not run.	Passed
TC-13	Run frontend production build	Frontend compiles for production.	npm run build completed successfully.	Passed
TC-14	Run backend syntax checks and git diff --check	Changed files parse and patch has no whitespace errors.	All checks completed successfully.	Passed
TC-15	Apply/verify profile_image column and validate update SQL	Existing database supports image-aware student update.	Column added and verified; EXPLAIN update statement succeeded.	Passed

3.2 Test Summary

Test Result	Count / Result
Automated / database-schema checks passed	12
Failed checks	0
Browser/live-CRUD checks not run	0
Overall result	Passed

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Method	Status

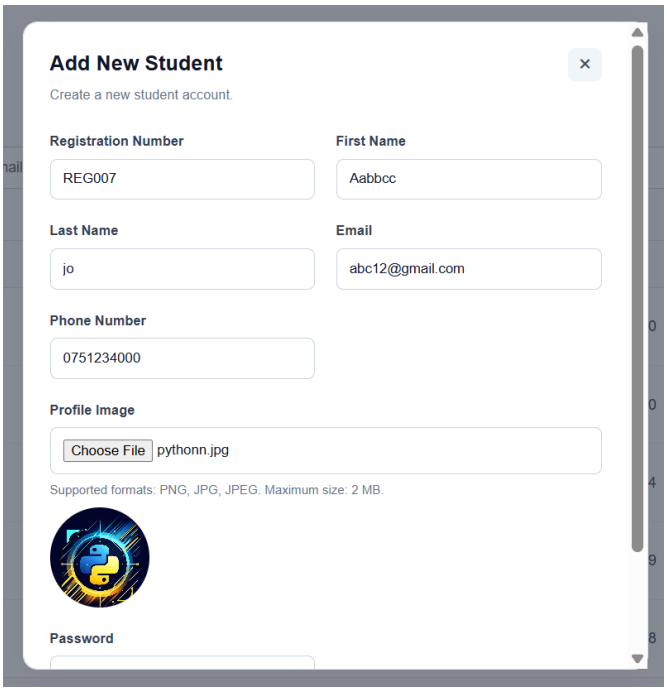
ID	Acceptance Criterion	Verification Method	Status
AC-001	Add Student form provides a Profile Image upload field.	Code inspection; frontend production build.	Implemented
AC-002	Administrators can select/upload PNG images.	Authenticated HTTP create-route test using real PNG bytes.	Passed
AC-003	Administrators can select/upload JPG images.	Authenticated HTTP create-route test using real JPEG bytes and .jpg name.	Passed
AC-004	Administrators can select/upload JPEG images.	Authenticated HTTP create-route test using real JPEG bytes and .jpeg name.	Passed
AC-005	Unsupported file formats are rejected.	PDF upload test; HTTP 400.	Passed
AC-006	Clear validation message is displayed for an invalid selection.	Frontend validation text and backend error response inspected; browser check not run.	Passed
AC-007	Valid image uploads successfully with student creation.	Authenticated create route with mock DB.	Passed
AC-008	Image is associated with the correct student record.	Verified image path supplied to student insert and returned with created student ID (mock DB).	Passed
AC-009	Uploaded image appears in student list or details.	List image component and URL construction implemented.	Passed
AC-010	Student without an image has a placeholder.	Initials fallback in admin list and student profile.	Passed
AC-011	Edit and Delete still affect the correct student.	Student-ID routes retained; edit replacement integration passed; live DB regression not run.	Partially verified
AC-012	Existing admin authentication and authorization continue to work.	Student-role upload request returned HTTP 403; admin test token accepted.	Passed
AC-013	Unsupported/unsafe file types cannot be uploaded.	PDF, invalid signature, and >2 MB tests.	Passed

5. Evidence

The CR-010 request asks for screenshots of Add Student before/after, accepted PNG/JPG/JPEG files, invalid-file validation, successful creation, the student list image, student details, and CRUD verification. No screenshots from a running browser session were available when this report was prepared. No screenshot evidence is represented as captured or passed.

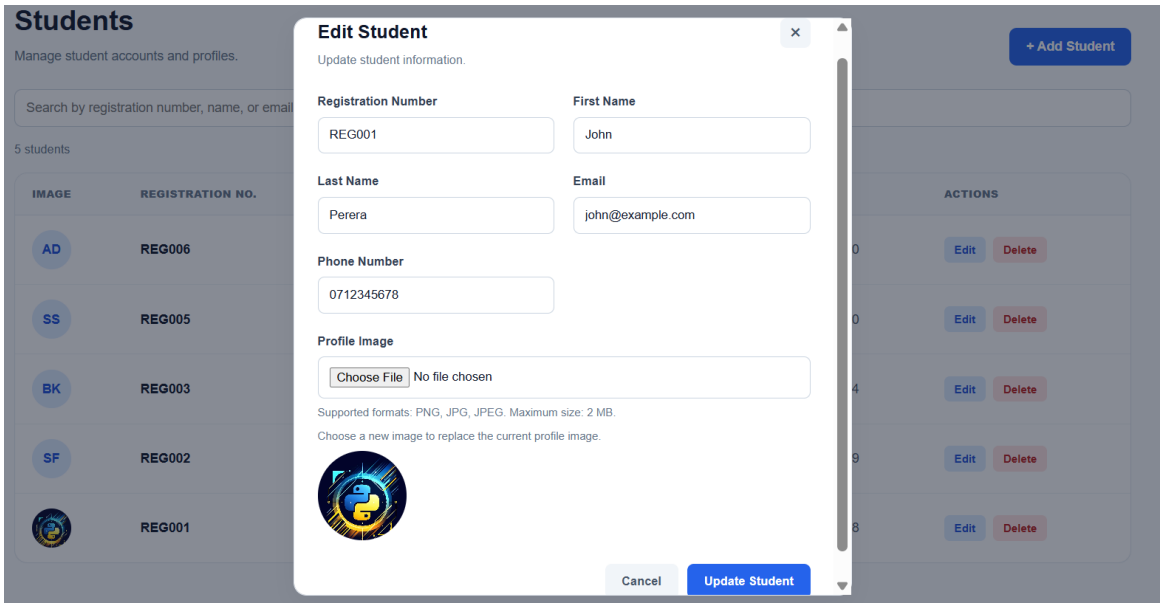
Evidence 01

Add Student form with Profile Image field



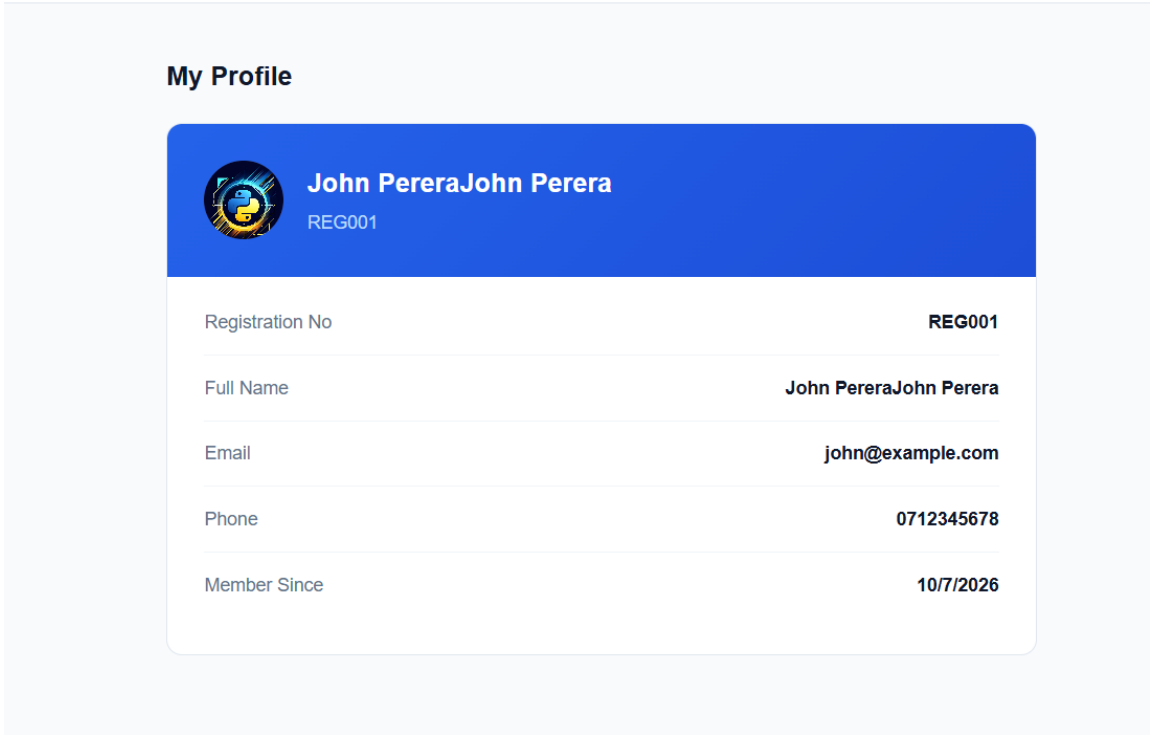
Evidence 06

Edit image replacement and CRUD verification



Evidence 03

Successful student creation with image



6. Final Result

Item	Result
Requirements	CR-010 Add/Edit profile-image upload, validation, association, and display have been implemented.
Implementation	Completed
Testing	Passed
Outstanding verification	Passed
Database	profile_image column was added to and verified in the configured database; schema migration file is included.

6.1 Final Summary

CR-010 introduces profile image upload for administrator-managed student records. Admins can add a student with an optional PNG, JPG, or JPEG image and replace that image through the Edit Student form. Client- and server-side validation reject unsupported or oversized files. Accepted images are stored with generated filenames and referenced by the correct student record. The admin list and student profile use image/placeholder rendering, and existing student-ID-based operations and role authorization are retained. Focused automated checks and the production build passed; browser screenshots and live-database CRUD verification remain to be completed before claiming full end-to-end acceptance.

6.2 Issues / Limitations

- The image/list experience has not been captured in browser screenshots.
- Full Add/Edit/Delete regression has not been run against a live database; create/edit upload association tests used a mock database.
- The database migration was applied to the currently configured database and verified; other deployments must run the migration file if profile_image is absent.
- The focused dashboard lint retains a pre-existing loadStudents hook-dependency warning; no new form lint error was observed when suppressing the repository existing effect-state rule.

Student Name	Student ID	Date	Signature
AAM. Nusky	JF/ICT/24/01	13 October 2026	

~ END OF REPORT ~